

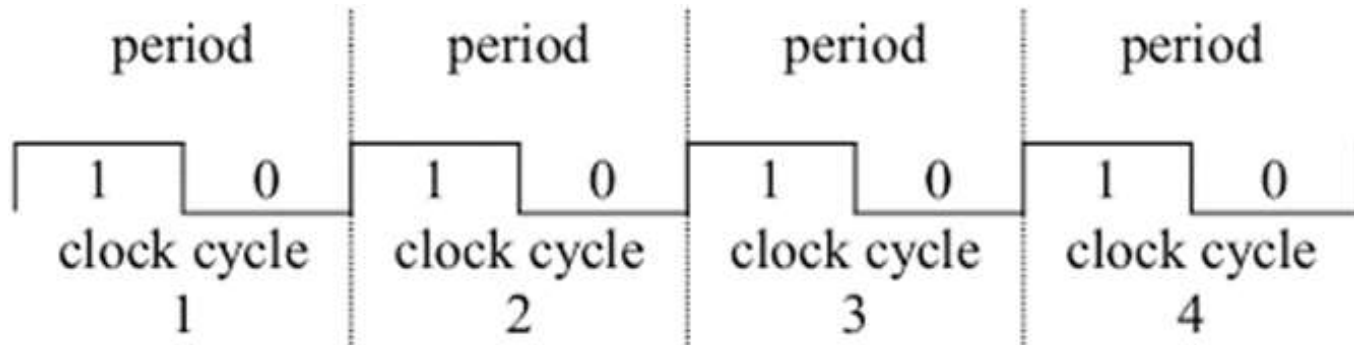
Temel Konular ve ARM

Hafta-2

Bu Derste neler öğrenilecek

- ▶ Mikroişlemci ve MCU'lar için bazı temel bilgileri hatırlayalım
- ▶ Cortex-M4
- ▶ Gerekli yazılım ve geliştirme ortamlarını tanıyalım

Frekans - Cycle Kavramları



Pipeline'ı hatırlayalım

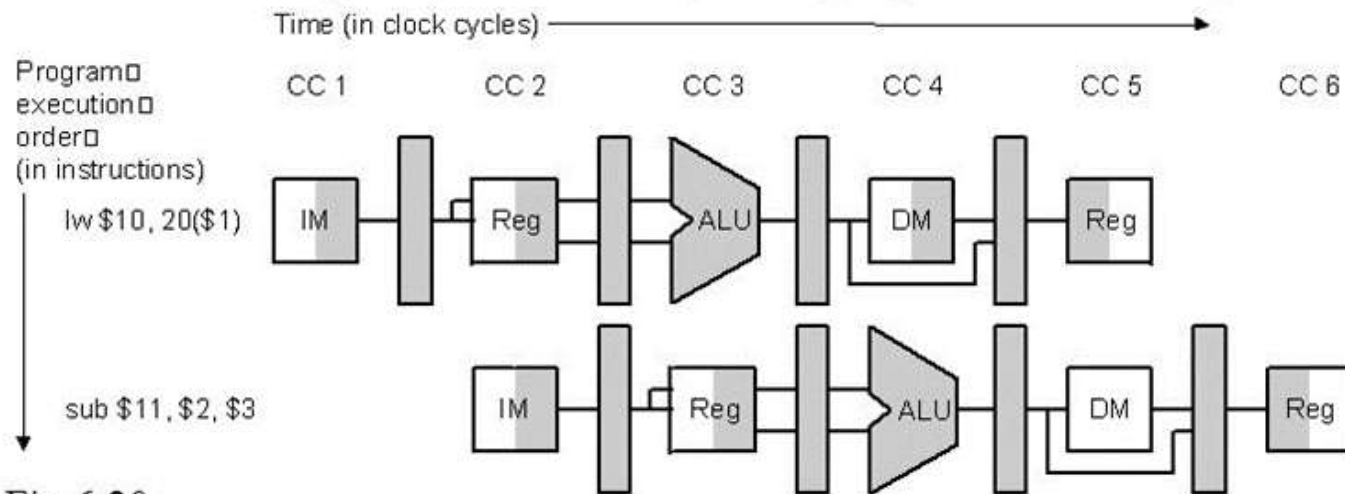
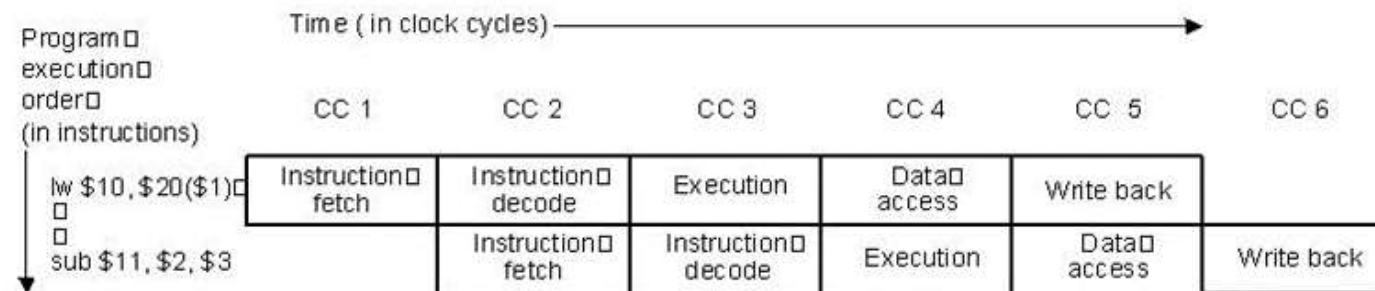


Fig 6.20



Sayı Sistemleri ve Dönüşümler

$$\begin{array}{cccccc} & C & D & 4 & .2 & 8 & C \\ & \swarrow & \swarrow & \swarrow & \swarrow & \swarrow & \swarrow \\ 1100 & 1101 & 0100 & 0010 & 1000 & 1100 \end{array}$$

$$\text{So, } (CD4.28C)_{16} = (110011010100.0010100011)_2$$

$$\begin{array}{r|l} 8090 & 16 \\ \hline 8080 & 505 \quad 16 \\ \hline 10 & 496 \quad 31 \quad 16 \\ (A) & 9 \quad 16 \quad 1 \\ & 15 \quad (F) \end{array}$$

$8090_{10} = 1F9A_{16}$

Dec	Hex
1	1
2	2
3	3
4	4
5	5
6	6
7	7
8	8
9	9
10	A
11	B
12	C
13	D
14	E
15	F

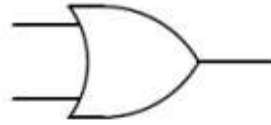
Decimal	Binary	Octal	Hexadecimal
0	0000	0	0
1	0001	1	1
2	0010	2	2
3	0011	3	3
4	0100	4	4
5	0101	5	5
6	0110	6	6
7	0111	7	7
8	1000	10	8
9	1001	11	9
10	1010	12	A
11	1011	13	B
12	1100	14	C
13	1101	15	D
14	1110	16	E
15	1111	17	F

Mantık Kapıları



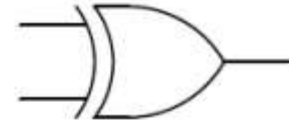
AND

A	B	Output
0	0	0
0	1	0
1	0	0
1	1	1



OR

A	B	Output
0	0	0
0	1	1
1	0	1
1	1	1



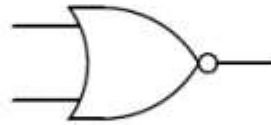
XOR

A	B	Output
0	0	0
0	1	1
1	0	1
1	1	0



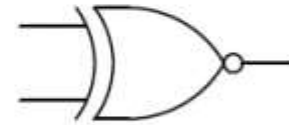
NAND

A	B	Output
0	0	1
0	1	1
1	0	1
1	1	0



NOR

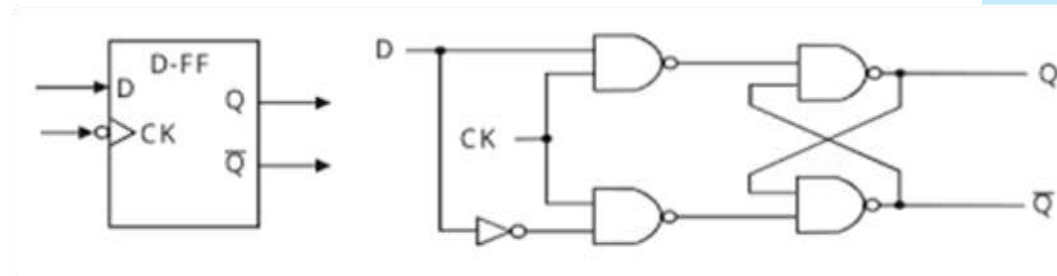
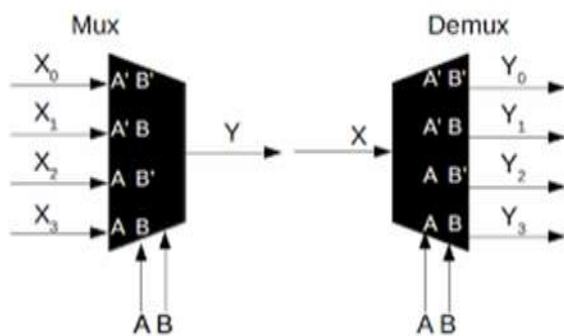
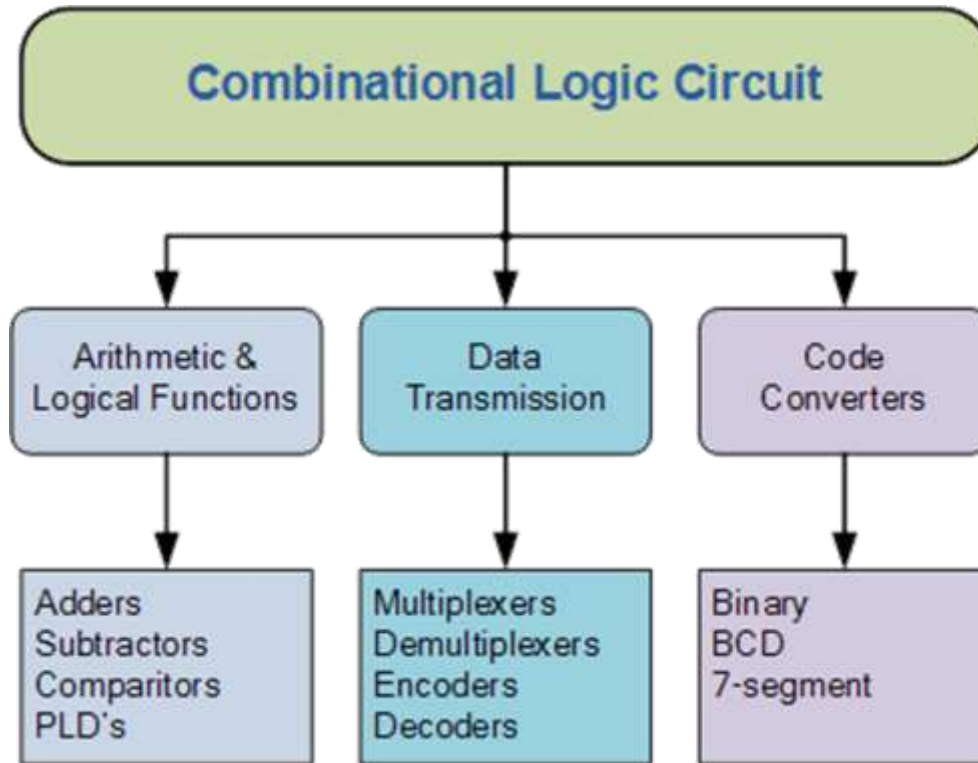
A	B	Output
0	0	1
0	1	0
1	0	0
1	1	0



XNOR

A	B	Output
0	0	1
0	1	0
1	0	0
1	1	1

Kombinasyonel ve Ardışıl Devreler



ASCII tablo niçin kullanılırdı? Daha gelişmiş kodlama teknolojisi neydi?

ASCII TABLE

Decimal	Hex	Char	Decimal	Hex	Char	Decimal	Hex	Char	Decimal	Hex	Char
0	0	[NULL]	32	20	[SPACE]	64	40	@	96	60	`
1	1	[START OF HEADING]	33	21	!	65	41	A	97	61	a
2	2	[START OF TEXT]	34	22	"	66	42	B	98	62	b
3	3	[END OF TEXT]	35	23	#	67	43	C	99	63	c
4	4	[END OF TRANSMISSION]	36	24	\$	68	44	D	100	64	d
5	5	[ENQUIRY]	37	25	%	69	45	E	101	65	e
6	6	[ACKNOWLEDGE]	38	26	&	70	46	F	102	66	f
7	7	[BELL]	39	27	'	71	47	G	103	67	g
8	8	[BACKSPACE]	40	28	(	72	48	H	104	68	h
9	9	[HORIZONTAL TAB]	41	29	)	73	49	I	105	69	i
10	A	[LINE FEED]	42	2A	*	74	4A	J	106	6A	j
11	B	[VERTICAL TAB]	43	2B	+	75	4B	K	107	6B	k
12	C	[FORM FEED]	44	2C	,	76	4C	L	108	6C	l
13	D	[CARRIAGE RETURN]	45	2D	-	77	4D	M	109	6D	m
14	E	[SHIFT OUT]	46	2E	.	78	4E	N	110	6E	n
15	F	[SHIFT IN]	47	2F	/	79	4F	O	111	6F	o
16	10	[DATA LINK ESCAPE]	48	30	0	80	50	P	112	70	p
17	11	[DEVICE CONTROL 1]	49	31	1	81	51	Q	113	71	q
18	12	[DEVICE CONTROL 2]	50	32	2	82	52	R	114	72	r
19	13	[DEVICE CONTROL 3]	51	33	3	83	53	S	115	73	s
20	14	[DEVICE CONTROL 4]	52	34	4	84	54	T	116	74	t
21	15	[NEGATIVE ACKNOWLEDGE]	53	35	5	85	55	U	117	75	u
22	16	[SYNCHRONOUS IDLE]	54	36	6	86	56	V	118	76	v
23	17	[ENG OF TRANS. BLOCK]	55	37	7	87	57	W	119	77	w
24	18	[CANCEL]	56	38	8	88	58	X	120	78	x
25	19	[END OF MEDIUM]	57	39	9	89	59	Y	121	79	y
26	1A	[SUBSTITUTE]	58	3A	:	90	5A	Z	122	7A	z
27	1B	[ESCAPE]	59	3B	;	91	5B	[	123	7B	{
28	1C	[FILE SEPARATOR]	60	3C	<	92	5C	\	124	7C	
29	1D	[GROUP SEPARATOR]	61	3D	=	93	5D	]	125	7D	}
30	1E	[RECORD SEPARATOR]	62	3E	>	94	5E	^	126	7E	~
31	1F	[UNIT SEPARATOR]	63	3F	?	95	5F	_	127	7F	[DEL]

ARM mimari profilleri



CPU architecture profiles

A-profile (Application)

Offers highest performance of all architecture profiles

[Learn More](#) >



R-profile (Real-Time)

Optimized for systems with real-time requirements

[Learn More](#) >



M-profile (Microcontroller)

Designed for small, low power, highly energy-efficient devices

[Learn More](#) >

Implementations

Arm Processor IP range:
[Cortex-A](#), [Neoverse](#), and
[Cortex-X](#)

Arm Processor IP range:
[Cortex-R](#)

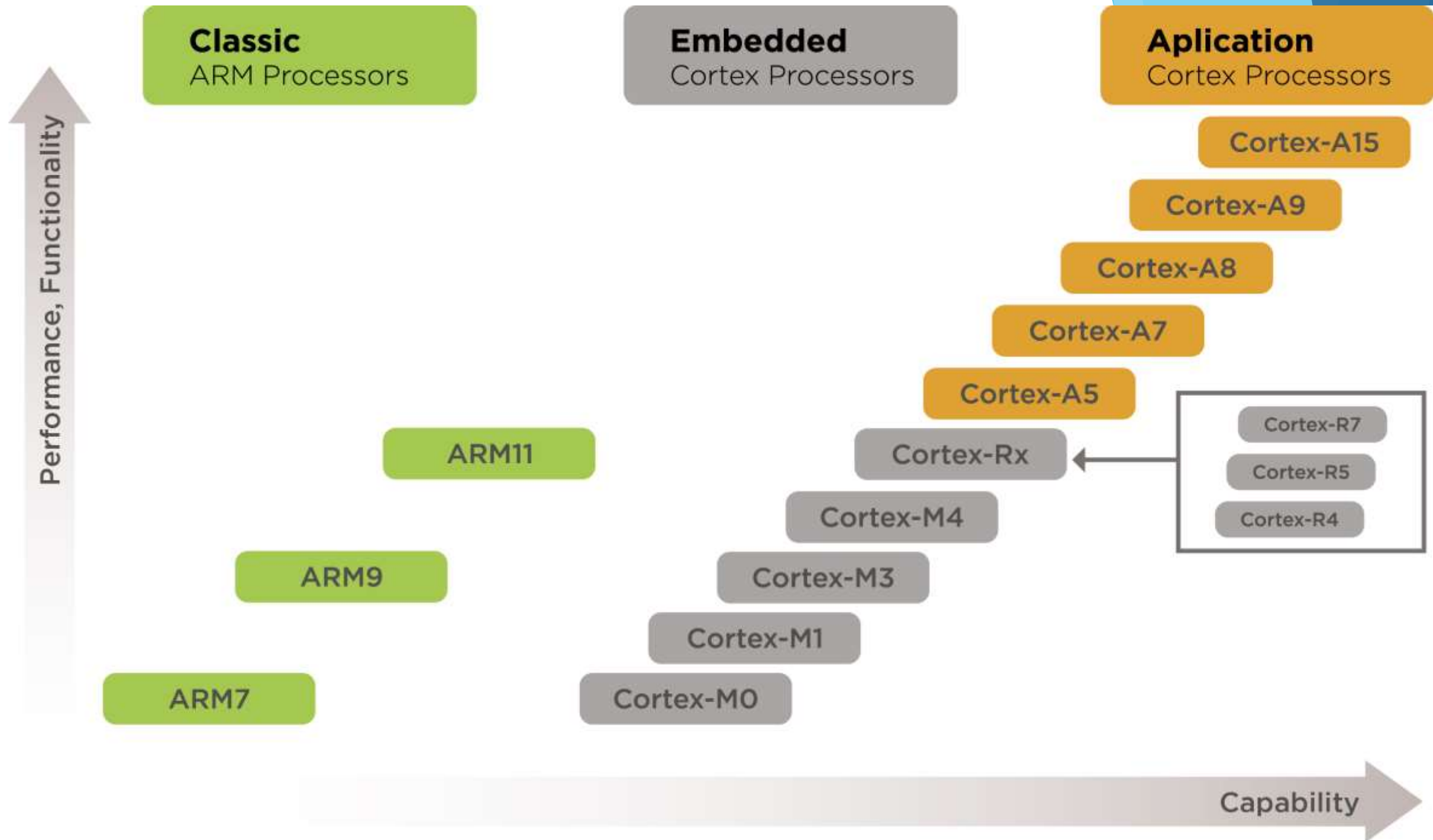
Arm Processor IP range:
[Cortex-M](#)

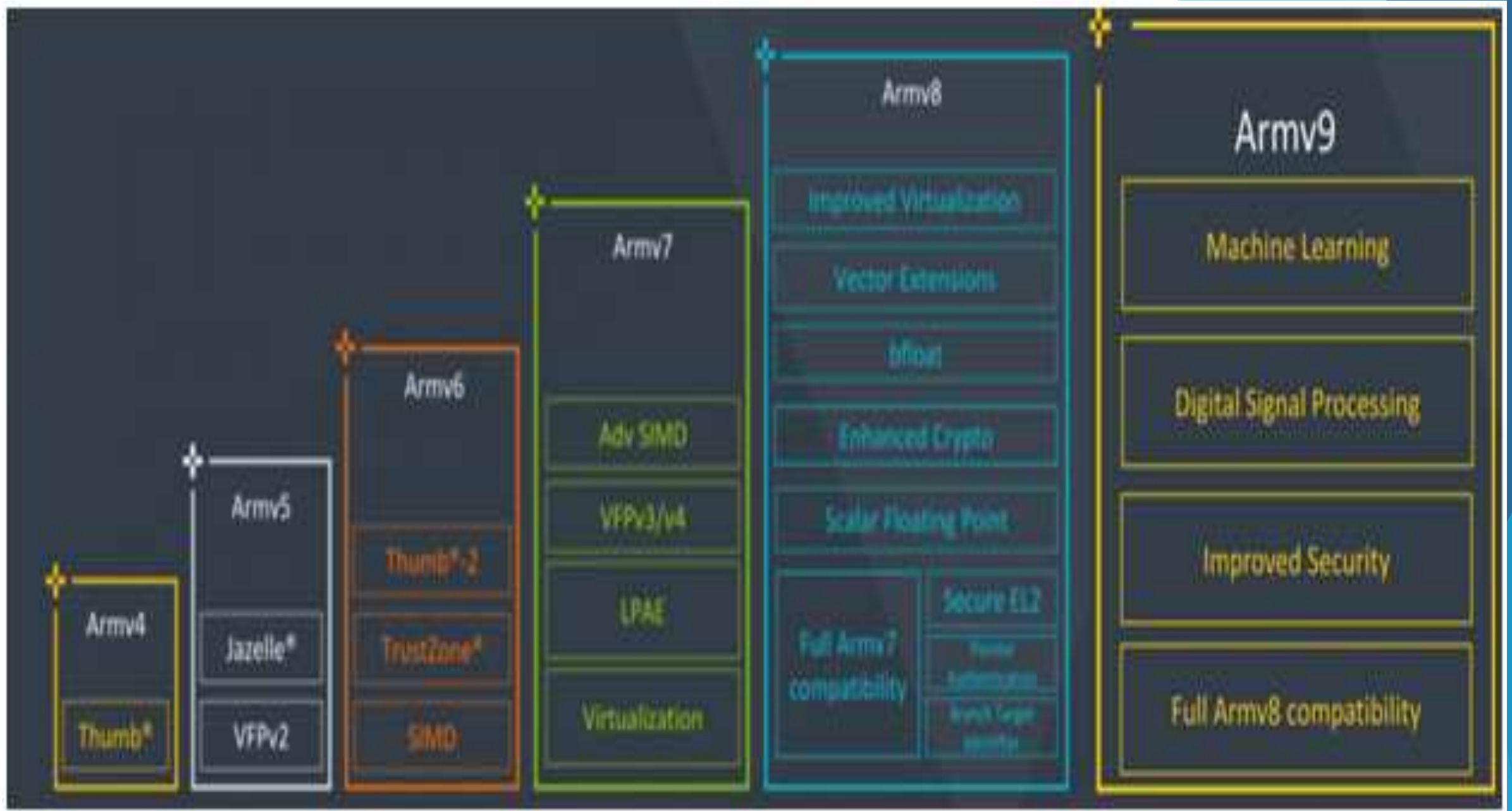
Features

- Offers highest performance of all architecture profiles

- Optimized for systems with real-time requirements

- Designed for small, low power, highly energy-efficient devices

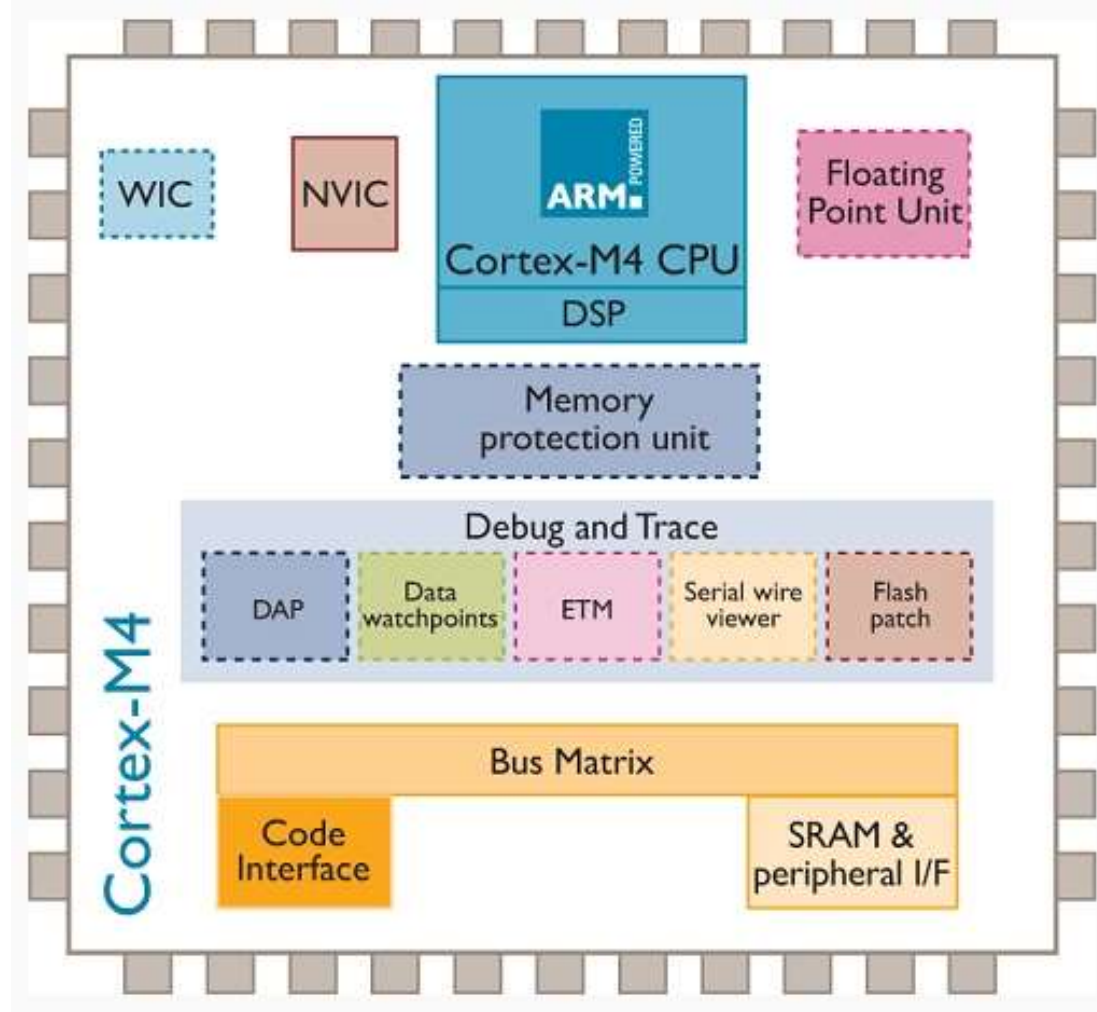




Ürün	Mimari (ISA)	Bazı İşlemciler	Özellik	Yıl
ARM1 - ... ARM11	ARMv1 ... Armv6	ARM600,ARM1156 vb.		1985-2005
SecurCore	ARMv6-ARMv7	SC100 - SC300		1998-2005
Cortex-M	ARMv7-M	Cortex-M3- M4 - M7	MCU	2003-2014
Cortex-M	ARMv8-M	Cortex -M23-M33-M85	MCU	2016-2022
Cortex-R	ARMv7-R	Cortex-R5 - R7 -R8	Real-time	2011-2016
Cortex-R	ARMv8-R	Cortex-R82	Real-time	2020
Cortex-A (32bit)	ARMv7-A	Cortex-A5-A7-A8-A12-A17	MPU	2009-2014
Cortex-A (64bit)	ARMv8-A (AArch64)	Cortex-A34-A55-A65-A76	MPU-Mobil	2016-2018
Cortex-A (64bit)	ARMv9-A	Cortex-A510-A710-A715	MPU-Mobil	2021
Cortex-X	ARMv8 - ARMv9	Cortex-X1-X2-X3	MPU-Mobil	2020-2022
Neoverse	ARMv8.2 -ARMv9	Neoverse V2- E2 - Neoverse-N2	MPU- Server	2021-2022

Cortex-M4 / STM32F4

- ▶ Core: 32-bit
- ▶ FPU desteği (IEEE-754 Single precision)
- ▶ 84MHz'e kadar frekans desteği
- ▶ 512KB'a kadar Flash
- ▶ 96KB'a kadar SRAM
- ▶ 1.7V ile 3.6V besleme ve I/O işlemleri
- ▶ POR/BOR desteği
- ▶ Dahili osilatör desteği
- ▶ 16 kanal 12 bit ADC
- ▶ 16-stream DMA
- ▶ 11 adet Timer
- ▶ 81 I/O port
- ▶ NVIC interrupt kontrolör
- ▶ I2C SPI ve USART desteği



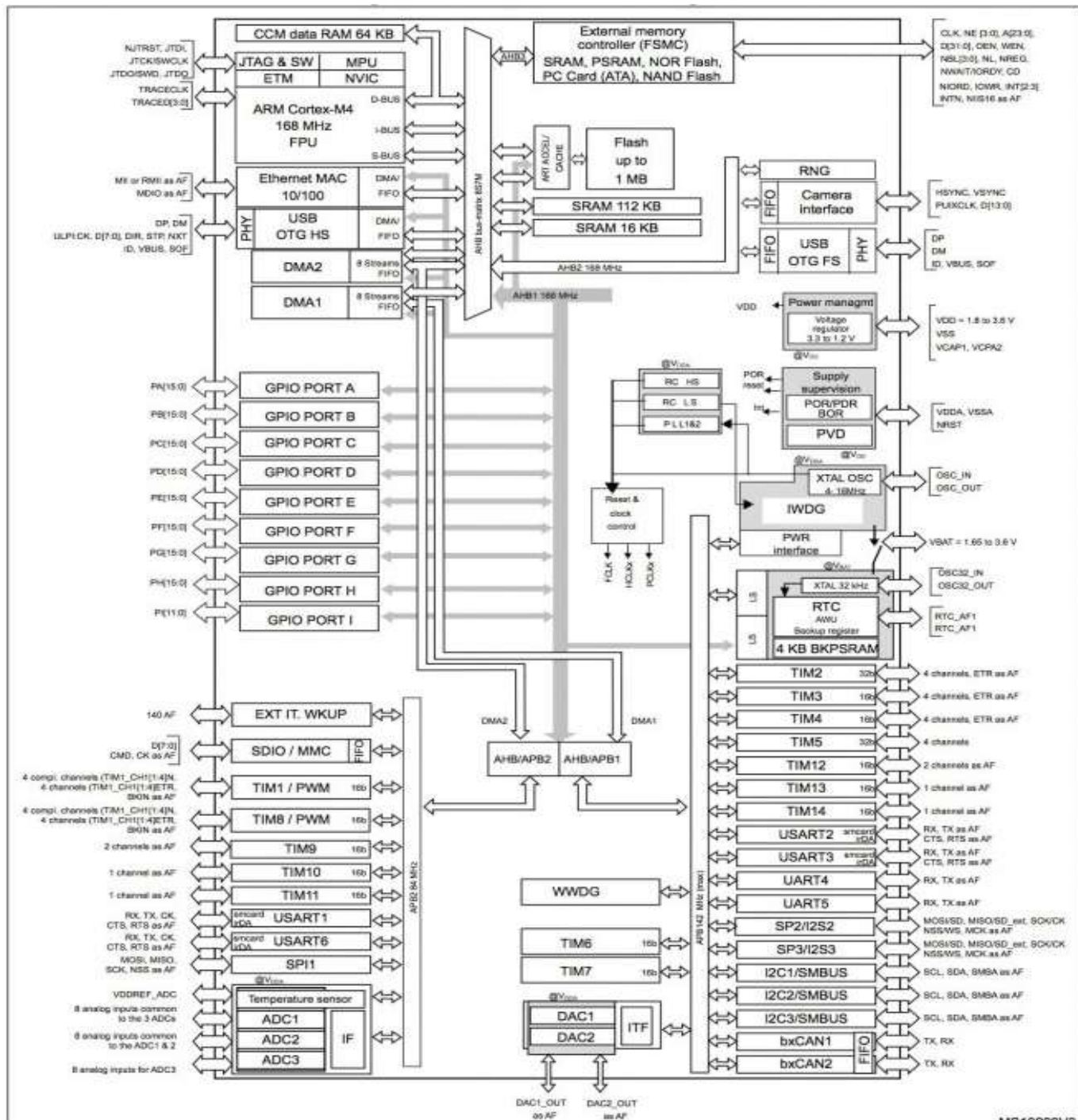
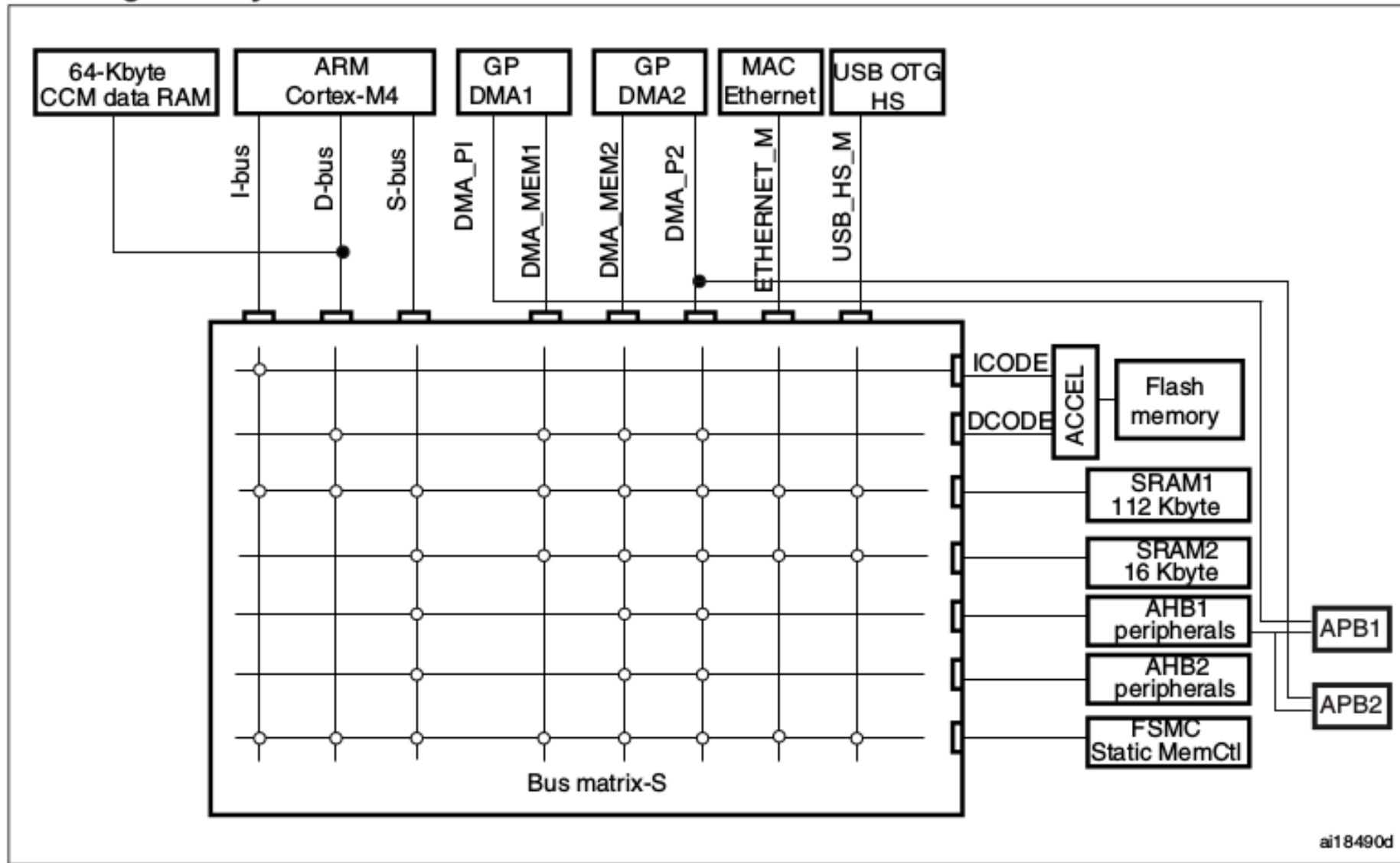


Figure 1. System architecture for STM32F405xx/07xx and STM32F415xx/17xx devices



Geliştirme Ortamları

- ▶ Keil
- ▶ Atollic Truestudio
- ▶ IAR





► Ders sonu